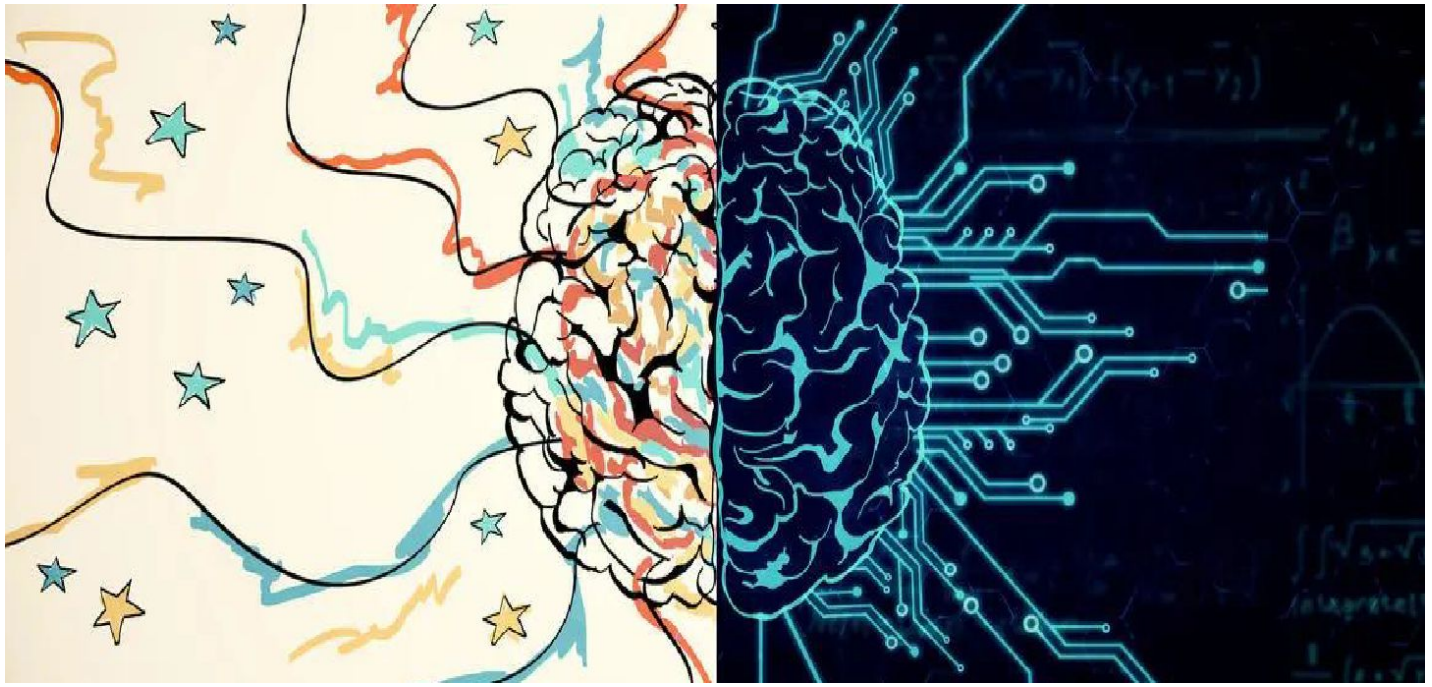
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AI Foundations for Engineers
Unit 1



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- What is AI?
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- The State of the Art
- Risks and Benefits of AI
- What is Responsible AI? Why it is important
- Intelligent Agents and Environments
- The concept of Rationality
- The Nature of Environments


spatial
visualizing the world in 3D
naturalist
understanding living things and reading nature
intra-personal
understanding yourself, what you feel, and what you want
musical
discerning sounds, their pitch, tone, rhythm, and timbre
THE TYPES OF
INTELLIGENCE
by Mark Vital
finding the right words to express what you mean
linguistic
quantifying things, making hypotheses and proving them
logical-mathematical
coordinating your mind with your body
bodily-kinesthetic
sensing people's feelings and motives
interpersonal
tackling the questions of why we live, and why we die
existential

How are humans intelligent?

- For thousands of years, we have tried to understand *how we think*; that is, how a mere handful of matter can perceive, understand, predict, and manipulate a world far larger and more complicated than itself.
- Learning • Reasoning • Problem Solving and Creativity • Social Behavior

- Experiencing our Environment with our senses:
  - Hearing
  - Sight
  - Touch
  - Taste
  - Smelling

Ways that People Think and Learn About Things

- If you have a problem, think of a past situation where you solved a similar problem.
- If you take an action, anticipate what might happen next.
- If you fail at something, imagine how you might have done things differently.
- If you observe an event, try to infer what prior event might have caused it.
- If you see an object, wonder if anyone owns it.
- If someone does something, ask yourself what the person's purpose was in doing that.

What is AI?

Artificial Intelligence (AI) is the branch of computer science that enables machines to perform tasks that normally require human intelligence such as:

- ❑ Learning
- ❑ Reasoning
- ❑ Problem-solving
- ❑ Perception
- ❑ Language understanding

Example:

- ❖ Voice assistants like **Alexa**
- ❖ Recommendation systems used by **Netflix**
- ❖ Self-driving technology by _____

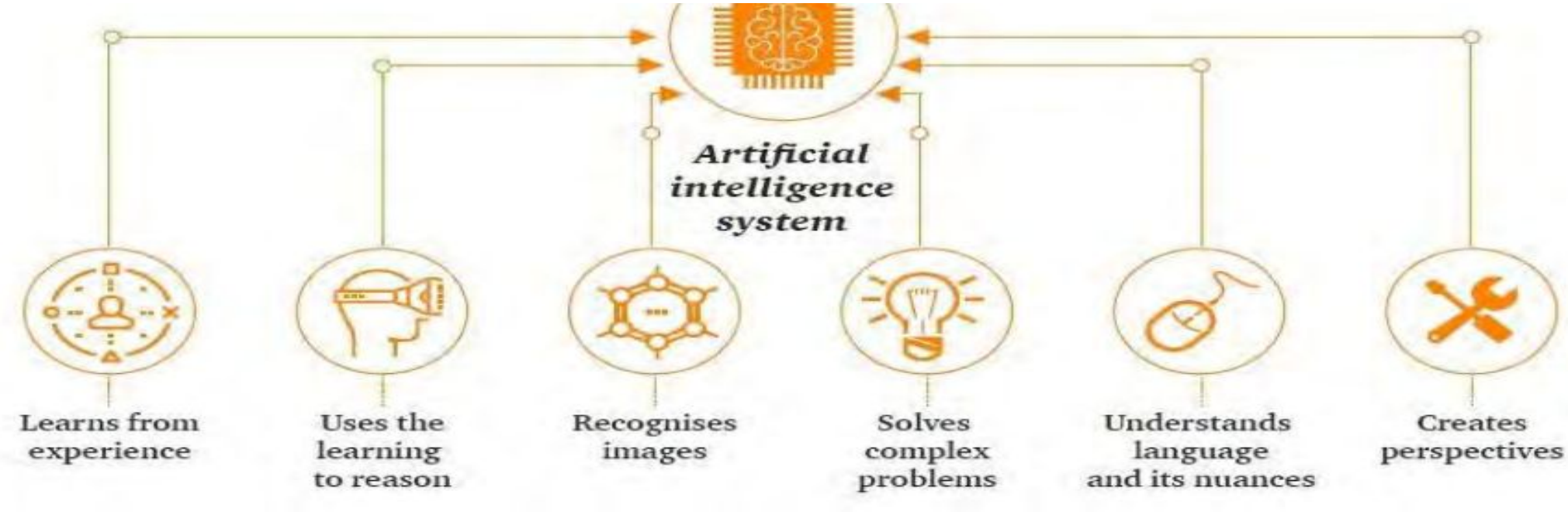
What is AI?

Artificial intelligence can be viewed from a variety of perspectives.

From the perspective of intelligence: A system with intelligence is expected to behave as intelligently as a human

From a business perspective: AI is a set of very powerful tools, and methodologies for using those tools to solve business problems.

From a programming perspective :AI includes the study of symbolic programming, problem solving, and search.



What is AI?

Healthcare AI Diagnosis System

A hospital uses AI to detect cancer from X-ray images.

- Input: Patient scan
- Process: AI model analyzes patterns
- Output: Early-stage tumor detection

Result: Faster diagnosis and improved survival rates

What is AI?

Thinking Humanly	Thinking Rationally
"The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense." (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Acting Humanly	**Acting Rationally**
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"Computational Intelligence is the study of the design of intelligent agents." (Poole *et al.*, 1998) "AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

Figure 1.1 Some definitions of artificial intelligence, organized into four categories.

Thinking Humanly: The Cognitive modeling Approach

If we are going to say that a given program thinks like a human, we must have some way of determining how humans think. We need to get *inside* the actual workings of human minds.

There are three ways of doing this:

1 Introspection

2 Psychological Experiments

3 Brain Imaging

Thinking Humanly: The Cognitive modeling Approach

Concept:

Machines think like humans by replicating human cognitive processes.

Foundation:

Based on psychology and neuroscience.

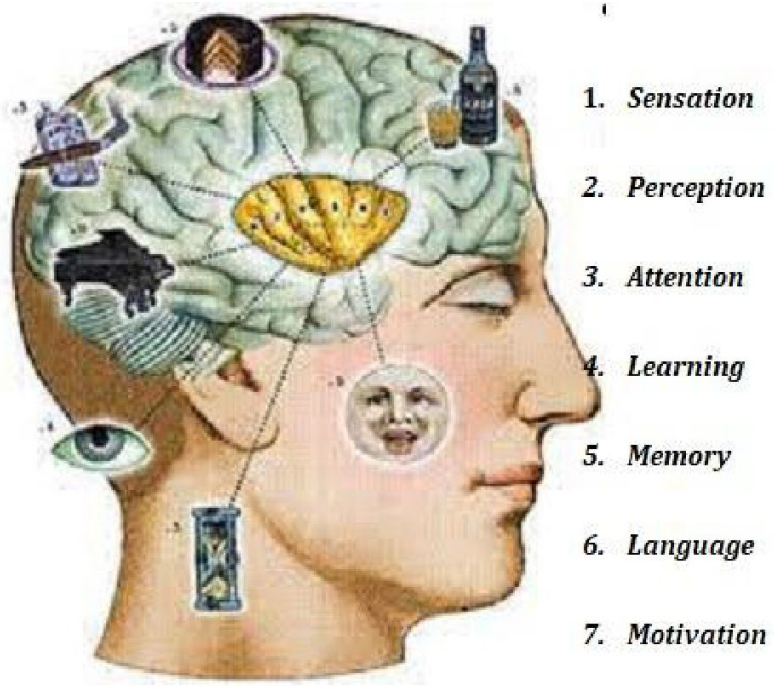
Example:

- Cognitive architectures like **ACT-R**

Case Study:

AI-based tutoring system:

- Observes how students solve math problems.
- Mimics human teacher thinking patterns.
- Adapts difficulty based on student mistakes



Mental Processes in Psychology

Acting Humanly – The Turing Test Approach

Concept:

A machine is intelligent if it can imitate human responses so well that a human cannot distinguish between the machine and a real human and was proposed by Alan Turing((1950)

Requirements:

- Natural Language Processing
- Knowledge Representation
- Automated Reasoning
- Machine Learning

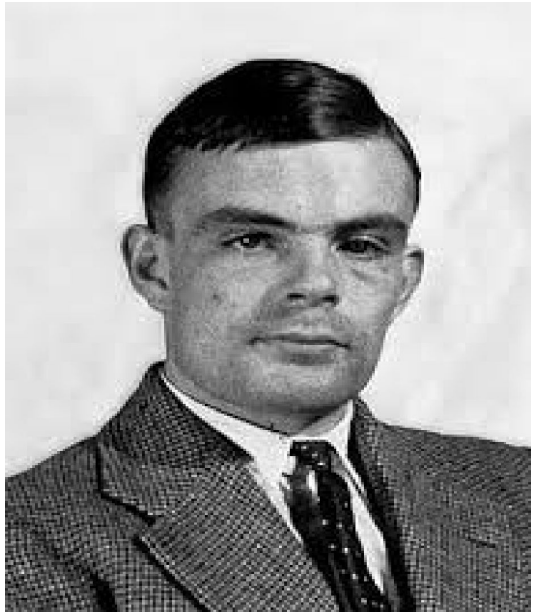
Example:

Chatbots like **ChatGPT**

Case Study:

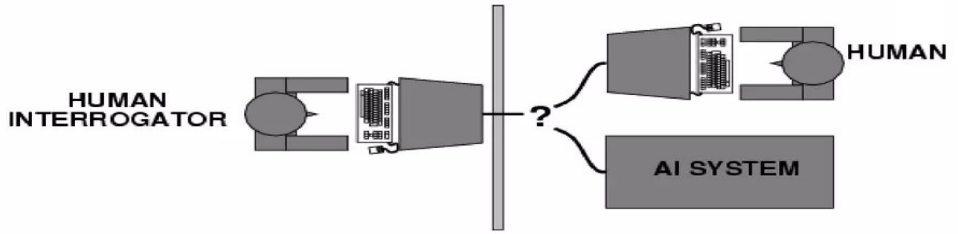
Customer service chatbot handling banking queries:

- 90% queries solved without human intervention.
- Customers unable to distinguish bot from human agent.



Acting humanly: Turing Test

- Turing (1950) "Computing machinery and intelligence":
- Operational test for intelligent behavior: the Imitation Game



- Interrogator asks questions of two “people” who are out of sight
- 30 minutes to ask whatever he or she wants
- Task: to determine only through the questions and answers which is which
- Computer deemed intelligent if interrogator can’t distinguish between person and computer.

Artificial intelligence is the enterprise of constructing an artificat that can pass the Turing text

Thinking Rationally – Laws of Thought Approach

Concept:

Based on logical reasoning.

AI systems follow formal logic to reach correct conclusions.

Example:

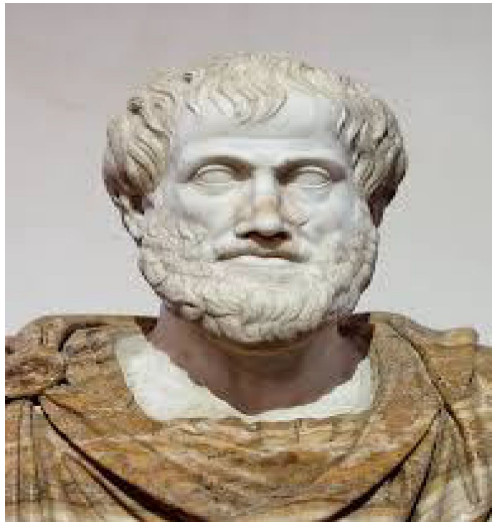
Expert Systems in medical diagnosis.

Case Study:

Medical expert system:

- Rule: IF fever AND cough THEN possible flu.
- Logical inference engine derives diagnosis.

Limitation: Real-world problems are uncertain and not always purely logical.



Thinking Rationally – Laws of Thought Approach

- Logic and laws of thought deals with studies of ideal or rational thought process and inference. The emphasis in this case is on the inferencing mechanism, and its properties. That is how the system arrives at a conclusion, or the reasoning behind its selection of actions is very important in this point of view. The soundness and completeness of the inference mechanisms are important here.
- The Greek philosopher Aristotle was one of the first to attempt to codify "right thinking," that is, irrefutable reasoning processes. His **syllogisms** provided patterns for argument structures that always yielded correct conclusions when given correct premises.

  for example, "Socrates is a man; all men are mortal; therefore, Socrates is mortal." These laws of thought were LOGIC supposed to govern the operation of the mind; their study initiated the field called **logic**.
- Logicians in the 19th century developed a precise notation for statements about all kinds of objects in the world and the relations among them. (Contrast this with ordinary arithmetic notation, which provides only for statements about *numbers*.)
- The so-called **logicist** tradition within artificial intelligence hopes to build on such programs to create intelligent systems.

Acting Rationally – Rational Agent Approach

Concept:

An agent acts to achieve the best expected outcome.

Rational Agent:

Perceives environment → Takes action → Maximizes performance measure.

Example:

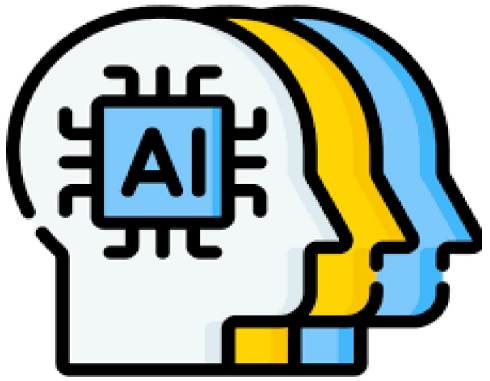
- Autonomous vehicles
- Chess-playing AI like **AlphaGo**

Case Study:

Self-driving car:

- Detects pedestrian.
- Calculates safest path.
- Brakes instantly.

Goal: Minimize accidents.



Acting Rationally – Rational Agent Approach

- The fourth view of AI is that it is the study of rational agents. This view deals with building machines that act rationally. The focus is on how the system acts and performs, and not so much on the reasoning process. A rational agent is one that acts rationally, that is, is in the best possible manner.
- An **agent** is just something that acts .All computer programs do something, but computer agents are expected to do more: operate autonomously, perceive their environment, persist over a prolonged time period, adapt to changes , and create and pursue goals.
  - A **rational agent** is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.
- In the "laws of thought" approach to AI, the emphasis was on **correct inferences**.
- Making correct inferences is sometimes *part* of being a rational agent, because one way to act rationally is to reason logically to the conclusion that a given action will achieve one's goals and then to act on that conclusion.
- On the other hand, correct inference is not *all* of rationality; in some situations, there is no provably correct thing to do, but something must still be done.

Guess What is used here?







Foundations of AI

Mathematics

AI depends on:

- Linear Algebra (Vectors, Matrices)
- Probability
- Statistics
- Calculus
- Optimization

Example:

Neural networks use matrix multiplication.

Case Study:

Fraud detection system uses probability to predict fraudulent transactions.

Foundations of AI

Economics

Concepts:

- Utility theory
- Decision theory
- Game theory

Example:

Online advertisement bidding systems.

Case Study:

E-commerce pricing model:

- Adjusts prices based on demand.
- Maximizes profit using economic models.

Foundations of AI

Neuroscience

Inspired by human brain structure.

Example:

Artificial Neural Networks (ANN)

Case Study:

Brain-inspired Deep Learning model for speech recognition.

Foundations of AI

Psychology

Studies:

- Learning behavior
- Memory
- Decision-making

Example:

Reinforcement learning based on reward system.

Case Study:

AI gaming agent:

- Receives reward for correct moves.
- Improves strategy over time.

Foundations of AI

Computer Engineering

Provides:

- Hardware (GPUs, TPUs)
- Parallel computing
- Embedded systems

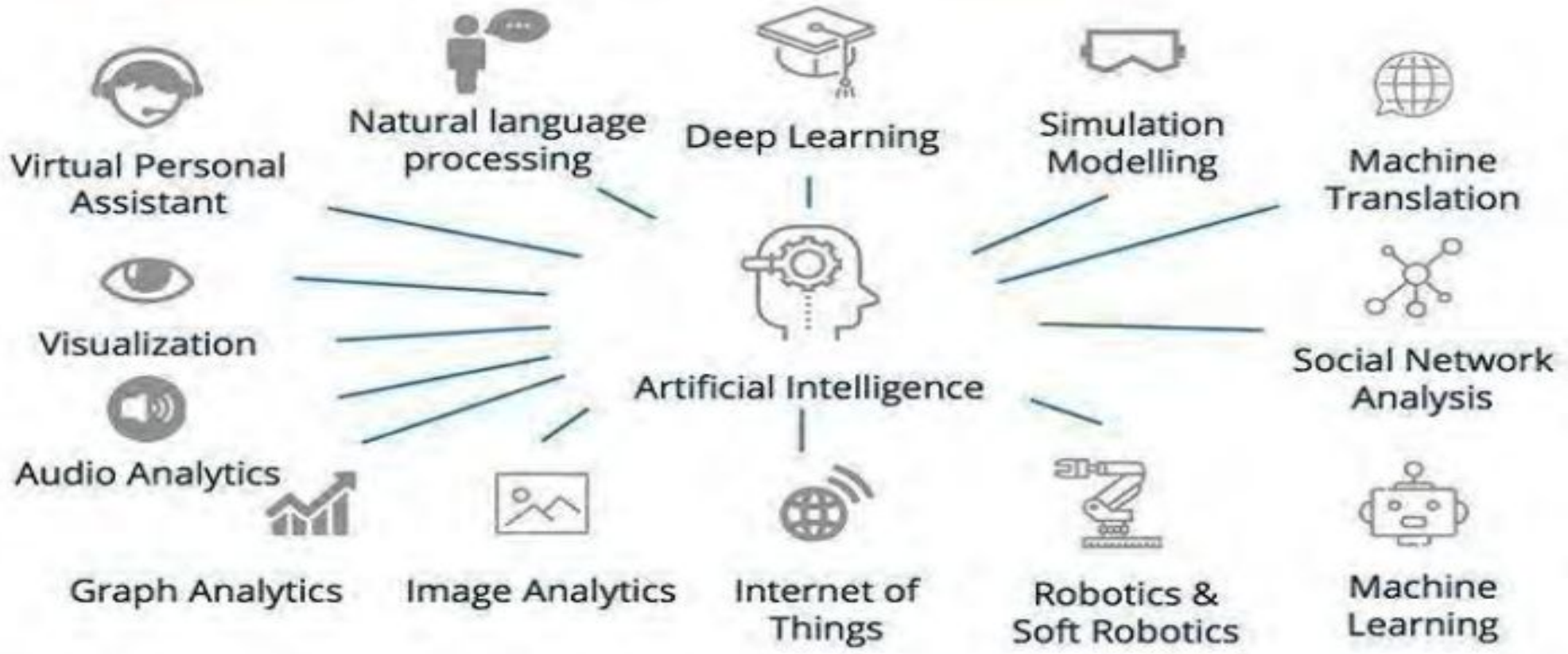
Example:

High-performance AI training using GPUs.

Case Study:

Smart surveillance system running on edge devices.

Possible applications for Artificial Intelligence



State of the Art in AI

Current Applications

- Generative AI (e.g., **ChatGPT**)
- Autonomous cars
- Medical AI
- AI in cybersecurity
- AI in education

Example:

Deep learning in image recognition.

Case Study:

AI detecting diabetic retinopathy with higher accuracy than human specialists.

Risks and Benefits of AI

Benefits

- Automation
- Improved accuracy
- 24/7 availability
- Cost reduction
- Better decision making

Case Study:

AI in agriculture predicting crop yield.

Risks

- Job displacement
- Bias and discrimination
- Privacy concerns
- Security threats
- Over-dependence

Case Study:

Facial recognition bias leading to incorrect identification.

Responsible AI

Responsible AI means designing and using AI systems in a way that is **fair, safe, transparent, and ethical**, so that the technology benefits people without causing harm.

In simple words, it means **AI should be used carefully and responsibly so that it treats everyone fairly and makes trustworthy decisions.**

Responsible AI

Responsible AI ensures systems are:

Fair : AI should treat all people equally and should not discriminate.
Transparent : The decisions made by AI should be understandable.
Accountable : Someone should be responsible for the AI system and its decisions.
Safe :AI systems should work correctly without causing harm.
Privacy-preserving : AI should protect people's personal data.

Responsible AI

Example 1: Loan Approval System

A bank uses AI to decide whether a person should get a loan.

- If the AI rejects people only based on gender or caste, it is **not responsible AI**.
- Responsible AI ensures the decision is based on **income, credit score, and repayment ability**, not personal bias.

Example 2: Medical Diagnosis

Hospitals use AI to detect diseases from medical images.

Example: AI used in healthcare systems like those developed by **IBM**.

Responsible AI ensures:

- Accurate diagnosis
- Doctors verify the AI results
- Patient data remains private

Example 3: Social Media Recommendation

Platforms like **YouTube** use AI to recommend videos.

Responsible AI should:

- Avoid promoting harmful or misleading content
- Protect users from misinformation

Why Responsible AI is Important?

- Ensure trust
- Legal compliance
- Ethical development
- Prevent harm

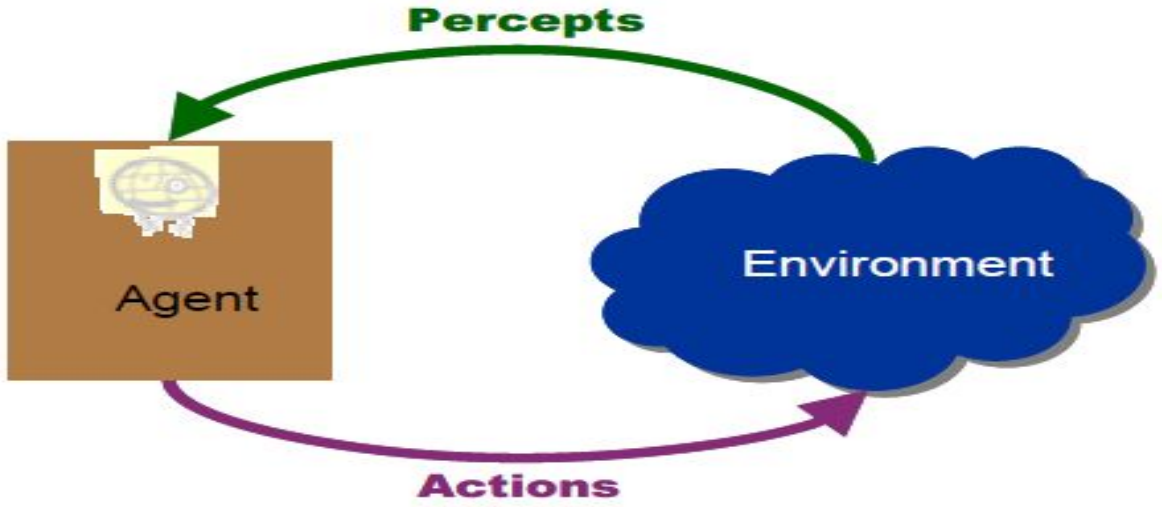
Intelligent Agents

Agent:

An **agent** is anything that **perceiving** its environment through **sensors** and **acting** upon that environment through **actuators**

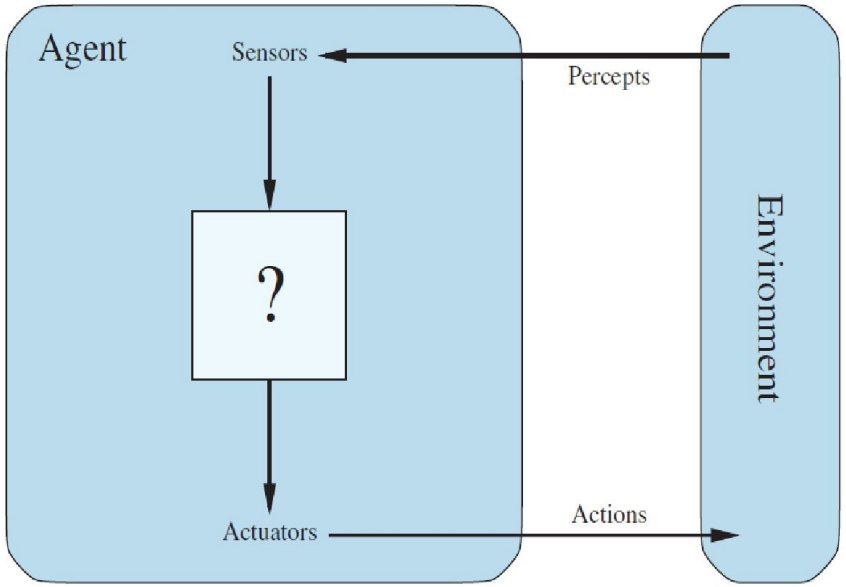
Example:

- Thermostat
- Self-driving car



Intelligent Agents

Agent X-thing, which perceives the environment through sensors and acts on the environment through actuators.



Human agent/Software agent/Robotic agent

Percept: Is what the agent senses from environment
Percept Sequence: Complete history of sensing
Agent function: Maps percept sequence to action
Agent Program: Is the implementation of the Agent function.

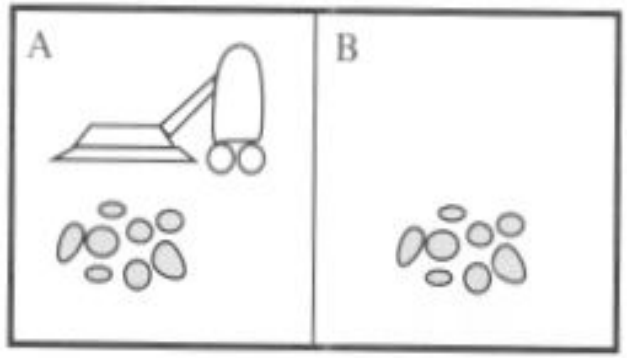
In general, *an agent's choice of action at any given instant can depend on the entire percept sequence observed to date, but not on anything it hasn't perceived.*

Intelligent Agents : Vacuum-cleaner world

Perception: Clean or Dirty? where it is in?

Actions: Move left, Move right, suck, do nothing

Percept sequence	Action
[*A*, *Clean*]	*Right*
[*A*, *Dirty*]	*Suck*
[*B*, *Clean*]	*Left*
[*B*, *Dirty*]	*Suck*
[*A*, *Clean*], [*A*, *Clean*]	*Right*
[*A*, *Clean*], [*A*, *Dirty*]	*Suck*
⋮	⋮
[*A*, *Clean*], [*A*, *Clean*], [*A*, *Clean*]	*Right*
[*A*, *Clean*], [*A*, *Clean*], [*A*, *Dirty*]	*Suck*
⋮	⋮



Function Reflex-Vacuum-Agent ([*location,status*]) return an action

If *status* = *Dirty* **then return** *Suck*

else if *location* = *A* **then return** *Right*

else if *location* = *B* **then return** *left*

Intelligent Agents : Vacuum-cleaner world

The obvious question, then, is this:

What is the right way to fill out the table?

In other words, what makes an agent good or bad, intelligent or stupid?

An **Intelligent Agent** must sense, must act, must be autonomous (to some extent),.

- It also must be rational.
- AI is about building rational agents.

Good Behaviour: The Concept of Rationality

Rational agent

- One that does the right thing
- = every entry in the table for the agent function is correct (rational).

What is correct?

- The actions that cause the agent to be most successful
- So we need ways to measure success.

Example:

Chess AI selecting best move.

Performance Measure

- When an agent is plunked down in an environment, it generates a sequence of actions according to the percepts it receives.
- This sequence of actions causes the environment to go through a sequence of states.
- If the sequence is desirable, then the agent has performed well.
- This notion of desirability PERFORMANCE is captured by a **performance measure** that MEASURE evaluates any given sequence of environment states.

Note: We said environment states not agent states, because if we define success in terms of agent's opinion ,then an agent could achieve perfect rationality simply by deluding itself that its performance was perfect.

Performance Measure

- Performance measure
  - An objective function that determines
    - How the agent does successfully
    - E.g., 90% or 30% ?
- An agent, based on its percepts
  - action sequence :

if desirable, it is said to be performing well.

- No universal performance measure for all agents

- A general rule:
  - Design performance measures according to
    - What one actually wants in the environment
    - Rather than how one thinks the agent should behave
- E.g., in vacuum-cleaner world
  - We want the floor clean, no matter how the agent behave
  - We don't restrict how the agent behaves
-

As a general rule, it is better to design performance measures according to what one actually wants in the environment, rather than according to how one thinks the agent should behave.

Rationality

What is rational at any given time depends on four things:

- The performance measure defining the criterion of success
- The agent's prior knowledge of the environment
- The actions that the agent can perform
- The agents's percept sequence up to now

Rational Agent

For each possible percept sequence,

- an rational agent should select
  - an action expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has

E.g., an exam

- Maximize marks, based on the questions on the paper & your knowledge

Omniscience

- An omniscient agent
  - Knows the *actual* outcome of its actions in advance and can act accordingly;
  - No other possible outcomes
  - However, impossible in real world
- An example

  crossing a street but died of the fallen cargo door from 33,000ft □ irrational?

- Based on the circumstance, it is rational.
- As rationality maximizes
  - *Expected* performance
- Perfection maximizes
  - *Actual* performance
- Hence rational agents are not ***omniscient***. Rationality maximizes *expected* performance, while perfection maximizes *actual* performance.

Information Gathering and Exploration

- Doing actions *in order to modify future percepts*—sometimes called **information gathering**—is an important part of rationality

A part from information gathering, agents must be allowed for **EXPLORATION**
Eg:example of information gathering is provided by the **exploration** that must be undertaken by a vacuum-cleaning agent in an initially unknown environment.

Learning

- Our definition requires a rational agent not only to gather information but also to **learn** as much as possible from what it perceives.
- The agent's initial configuration could reflect some prior knowledge of the environment, but as the agent gains experience this may be modified and augmented.
- Does a rational agent depend on only current percept?
  - No, the past percept sequence should also be used
  - This is called **learning**
  - After experiencing an episode, the agent
    - should adjust its behaviors to perform better for the same job next time.

Autonomy

- If an agent just relies on the prior knowledge of its designer rather than its own percepts then the agent lacks ***autonomy***

  A rational agent should be autonomous- it should learn what it can to compensate for partial or incorrect prior knowledge.

- E.g., a vacuum-cleaning agent that learns to foresee where and when additional dirt will appear will do better than one that does not.

Nature of Environments

Environment types:

1. Fully observable vs Partially observable
2. Deterministic vs Stochastic
3. Episodic vs Sequential
4. Static vs Dynamic
5. Discrete vs Continuous

Nature of Environments

Specifying the task environment

PEAS (Performance, Environment, Actuators, Sensors)

Before designing the agent, the PEAS description is the first step

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits, minimize impact on other road users	Roads, other traffic, police, pedestrians, customers, weather	Steering, accelerator, brake, signal, horn, display, speech	Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

Write PEAS for..

Agent: Social Media Recommendation System
Task: Maximize user engagement
Performance: ?
Environment: ?
Actuators: ?
Sensors: ?

Write PEAS for..

Agent: Social Media Recommendation System
Task: Maximize user engagement
Performance: Time spent on platform, clicks, likes, shares
Environment: Users, their devices, content feeds, user behavior
Actuators: Content ranking, feed updates, push notifications
Sensors: User interaction logs, click-through rates, scrolling behavior

Write PEAS for..

Agent: AI Teacher Bot
Task: Maximize student scores
Performance: ?
Environment: ?
Actuators: ?
Sensors: ?

Nature of Environment

Properties of the task environment

The dimensions of the task environment decide the type of agent design

Identify the following task environments on your own.



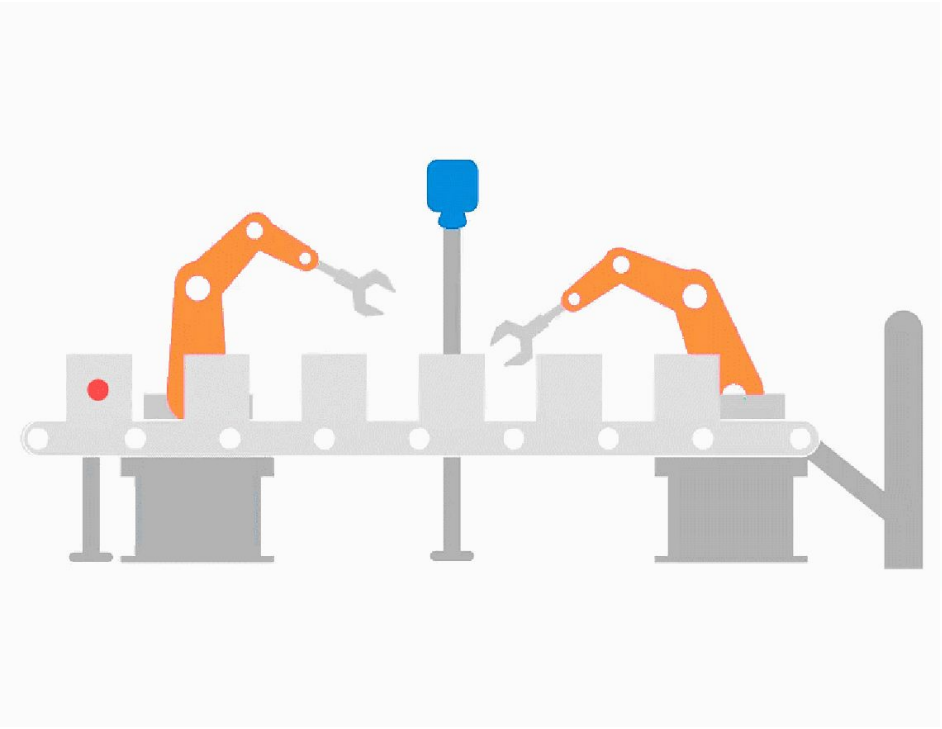
Property	Type/Description
Observability	**Partially Observable** – cannot see under furniture or detect all dirt directly
Agents	**Single Agent** – one robot acting independently
Deterministic vs Stochastic	**Stochastic** – dirt and obstacle placement may vary randomly
Episodic vs Sequential	**Sequential** – each action affects future performance (e.g., path, battery)
Static vs Dynamic	**Dynamic** – humans/pets may move objects while cleaning
Discrete vs Continuous	**Continuous** – movement, space, and time are continuous

Nature of Environment



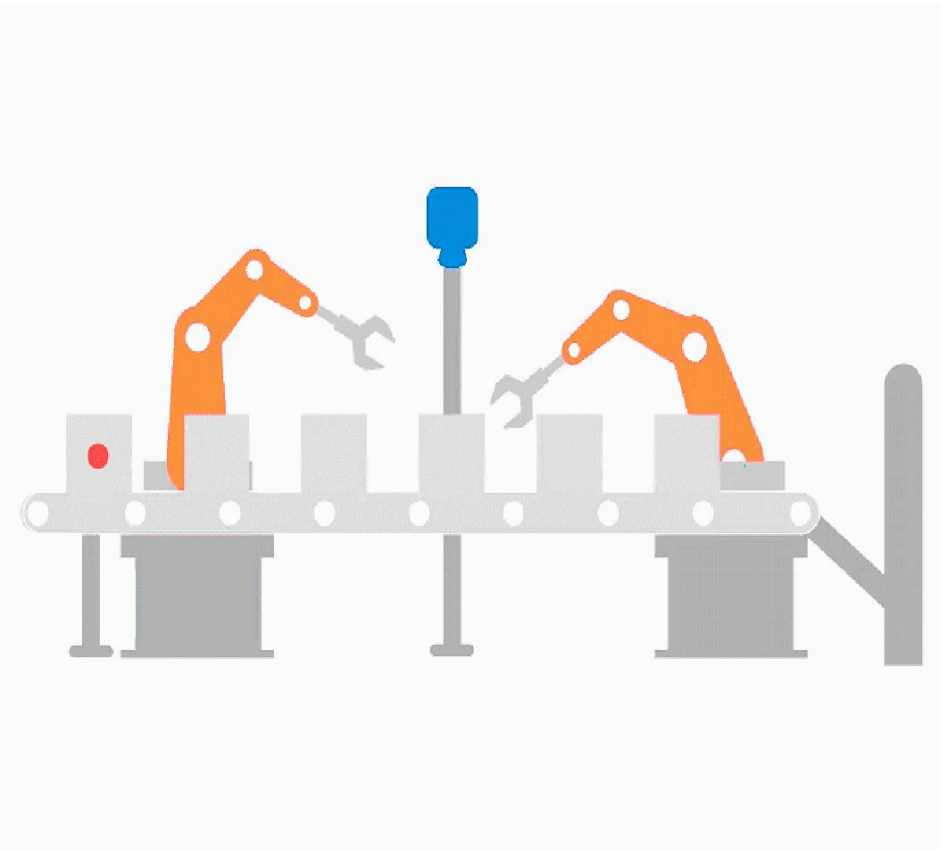
Property	Type/Description
Observability	Full/Partial ??
Agents	Single Agent / Multi-agent ??
Deterministic vs Stochastic	Stochastic / Deterministic ??
Episodic vs Sequential	Sequential/Episodic ??
Static vs Dynamic	Dynamic/Static ??
Discrete vs Continuous	Continuous/Discrete ??

Nature of Environment



Property	Type/Description
Observability	Full/Partial ??
Agents	Single Agent / Multi-agent ??
Deterministic vs Stochastic	Stochastic / Deterministic ??
Episodic vs Sequential	Sequential/Episodic ??
Static vs Dynamic	Dynamic/Static ??
Discrete vs Continuous	Continuous/Discrete ??

Nature of Environment



Feature	Type of Environment
Observability	Partially Observable – Not all defects may be visible at once; limited by camera angle, lighting, or occlusion.
Agents	Single Agent – One AI system performing inspection (though multiple instances may be deployed).
Deterministic vs Stochastic	Stochastic – Sensor noise, variations in lighting, speed of conveyor belt, or part placement can introduce randomness.
Episodic vs Sequential	Episodic – Each part inspection is independent of the previous one; decision does not affect future inputs.
Static vs Dynamic	Semi-Dynamic – Environment may change slowly (e.g., lighting shifts, belt speed), but parts move continuously.
Discrete vs Continuous	Continuous – Visual data and inspection space are continuous; possible discrete actions (accept/reject).

Nature of Environment

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

Figure 2.5 Examples of agent types and their PEAS descriptions.

Properties of Task Environment

1. Fully observable vs. Partially observable
2. Deterministic vs. stochastic
3. Episodic vs. sequential
4. Static vs. dynamic
5. Discrete vs. continuous
6. Single agent VS. multiagent
7. Known vs. unknown

Examples of Task Environment

Task Environment	Observable	Deterministic	Episodic	Static	Discrete	Agents
Crossword puzzle	Fully	Deterministic	Sequential	Static	Discrete	Single
Chess with a clock	Fully	Strategic	Sequential	Semi	Discrete	Multi
Poker	Partially	Strategic	Sequential	Static	Discrete	Multi
Backgammon	Fully	Stochastic	Sequential	Static	Discrete	Multi
Taxi driving	Partially	Stochastic	Sequential	Dynamic	Continuous	Multi
Medical diagnosis	Partially	Stochastic	Sequential	Dynamic	Continuous	Single
Image-analysis	Fully	Deterministic	Episodic	Semi	Continuous	Single
Part-picking robot	Partially	Stochastic	Episodic	Dynamic	Continuous	Single
Refinery controller	Partially	Stochastic	Sequential	Dynamic	Continuous	Single
Interactive English tutor	Partially	Stochastic	Sequential	Dynamic	Discrete	Multi

Figure 2.6 Examples of task environments and their characteristics.

Thank You!